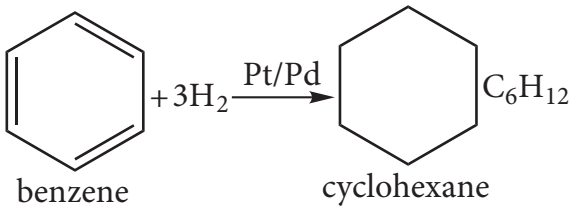
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(ii) Addition Reaction:

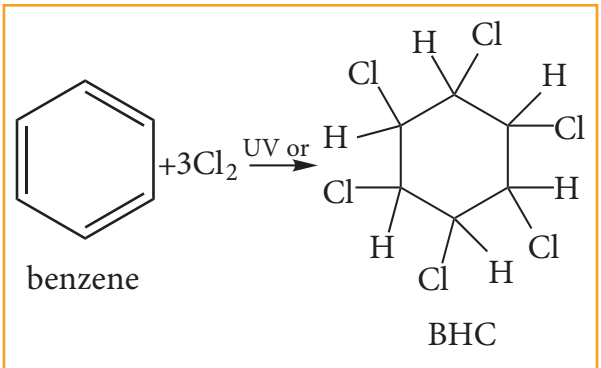
a. Hydrogenation of benzene:

Benzene reacts with hydrogen in the presence of Platinum or Palladium to yield Cyclohexane. This is known as hydrogenation.



b. Chlorination of Benzene:

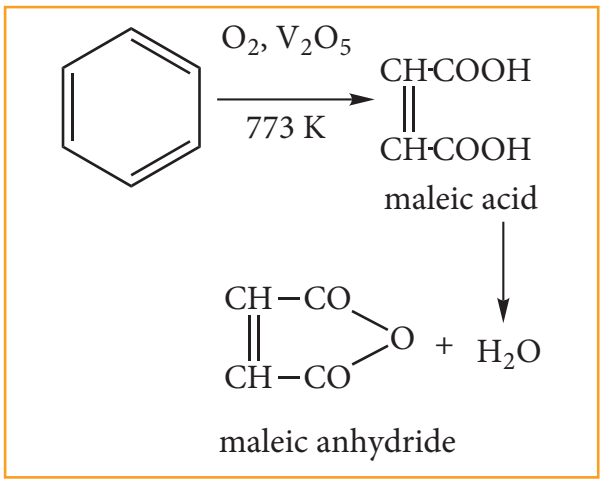
Benzene reacts with three molecules of Cl_2 in the presence of sun light or UV light to yield Benzene Hexa Chloride (BHC) $C_6H_6Cl_6$. This is known as gammaxane or Lindane which is a powerful insecticide.



(Iii) Oxidation:

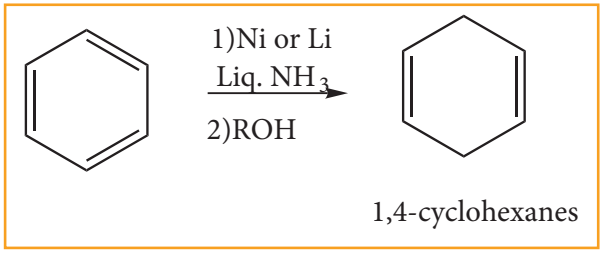
a. Vapour – phase oxidation:-

Although benzene is very stable to strong oxidizing agents, it quickly undergoes vapour phase oxidation by passing its vapour mixed with oxygen over V_2O_5 at 773k. The ring breaks to give maleic anhydride.



b. Birch reduction:

Benzene can be reduced to 1, 4-cyclohexadiene by treatment with Na or Li in a mixture of liquid ammonia and alcohol. It is the convenient method to prepare cyclic dienes.



Evaluate Yourself

24) Convert Ethyne to Benzene and name the process.

13.5.1 directive influence of a functional group in monosubtituted benzene:

When mono substituted benzene undergoes an electrophilic substitution reaction, the rate of the reaction and the site of attack of the incomeing electrophile depends on the functional group already attached to it. Some groups increase the reactivity of benzene ring and are known as activating groups. While others which decrease the reactivity are known as deactivating groups. We further divide